GENERATION 2

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ABSTRACT

Generation II is an interactive electronic music performance between a live improviser and the 'Evol' algorithmic music system. The work explores the premise that musical structure can arise from variations and elaboration of musical motifs. The work explores the automation of these process in the iterative improvised context of performance with a live algorithm.

This work continues the artist's exploration of performing with reflexive music software where the musician's performance is captured and fragments are transformed and re-used as part of a generative musical accompaniment (Gifford and Brown 2011; Brown, Gifford and Voltz 2016). In particular, the Evol system transforms the captured fragments over time, using them as motifs on which to elaborate. In Generation II multiple software agents using the Evol system create a layered texture that transforms in subtle ways over time. The agents are synchronised by an underlying time base. The effect is often reminiscent of the works of minimalist compositions from the 20th century.

Interaction between the performer and algorithmic music system is primarily via the performance itself. The Evol system includes real-time analysis of the performance and does feature extraction that guides the generative response. In addition, some direct performer control of the software agent behaviour is possible via a gestural interface.

The Evol system is MIDI-based and so the timbral character of the work is determined by sound choices in the playback engine. In the performance of *Generation I* these were tonal analogue synthesizer tones, whilst for Generation II these will be mixed with more aggressive glitch and noise based timbres.'

REFERENCES

Gifford, Toby, and Andrew R. Brown. "Beyond Reflexivity: Mediating between Imitative and Intelligent Action in an Interactive Music System." In *Proceedings of the 25th BCS Conference on Human-Computer Interaction*, edited by Katie Wilkie, Rose Johnson, and Simon Holland. Newcastle Upon Thyme, 2011.

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